

Koushick Padmanabhan

New Jersey

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[Github - Koushick Padmanabhan](#) | [Portfolio - Koushick Padmanabhan](#)

Education

Montclair State University

Master of Science, Computer Science

• **GPA:** 3.6/4.0

• **Coursework:** Software Engineering, Web Development, DBMS, Computer Algorithms and Analysis, Data Science

Sep 2024 - May 2026

Montclair, NJ

St. Joseph's College of Engineering

Bachelor of Technology, Electronics and Communication Engineering

• **GPA:** 3.7/4.0

Aug 2018 - Jun 2022

India

Professional Experience

Academy St. Thrift

Full Stack Developer Intern

- Developed and deployed a scalable full-stack web application on AWS using React, FastAPI, PostgreSQL, and Docker, supporting multi-user workflows and improving system reliability with 99%+ uptime.
- Built and integrated an EfficientNet-based image classification pipeline trained on 60K+ images, achieving 94%+ accuracy for footwear detection and reducing manual validation efforts by 40%.
- Developed a full-stack application using React, FastAPI, and PostgreSQL, integrating an LLM-powered image analysis pipeline to classify products and extract attributes (e.g., color, category), reducing manual verification effort by 45% and improving data consistency across listings.
- Containerized applications using Docker and implemented version control with Git, enabling seamless deployment and collaboration across team environments, reducing deployment inconsistencies by 30%.

Jan 2026 - May 2026

New Jersey, United States

Infosys

Software Development Engineer

- Designed and developed scalable backend services using Python (FastAPI) and RESTful APIs, handling 10K+ daily requests with optimized response times under 150ms.
- Migrated and modularized legacy application components into microservices architecture, improving system scalability and reducing deployment time using Docker and AWS services (EC2, S3).
- Implemented end-to-end features across the stack using React and backend APIs, enhancing user workflows and increasing feature adoption by 25%.
- Developed and integrated CI/CD pipelines using Git, GitHub Actions/Jenkins, enabling automated testing and deployments, reducing production defects by 40% and accelerating release cycles.

Aug 2022 - Aug 2024

Chennai, India

Retech Solutions Pvt. Ltd

Junior Developer Intern

- Assisted in developing and debugging web application features using HTML, CSS, JavaScript, and Python, resolving minor issues and improving code reliability, contributing to a 15% reduction in reported bugs during testing.
- Supported senior developers in testing and maintaining code, working with REST APIs and MySQL/PostgreSQL, and improving code readability and documentation, reducing onboarding time for new features by 10%.

Aug 2020 - Oct 2020

Chennai, India

Projects

Open Fiesta

- Built a multi-model AI chat platform enabling users to interact with 200+ AI models (OpenAI, Claude, Gemini, Grok, etc.) in parallel with real-time streaming, allowing side-by-side comparison of responses for improved decision-making.
- Developed a modern full-stack application using Next.js, TypeScript, PostgreSQL, and Vercel AI SDK, implementing authentication, persistent chat history, and configurable model parameters (temperature, top-p) for customizable AI interactions.

Sep 2025

OpenCut

- Built an end-to-end AI-powered video generation platform that transforms user prompts or URLs into fully generated short-form videos by integrating Vertex AI (Gemini), ElevenLabs, and Firecrawl for automated script, narration, and visual generation.
- Architected a scalable rendering pipeline using Remotion with a dedicated render worker (Bun + Hono), enabling real-time preview and export of videos as MP4s, with efficient storage and retrieval using Cloudflare R2.

Feb 2026

Customer-support-voice-agent

- Engineered a real-time AI-powered customer support voice agent by integrating LiveKit's WebRTC infrastructure with a FastAPI backend, enabling low-latency bidirectional audio streaming and automated token-based session authentication via a Next.js frontend API route.
- Designed and implemented a full-stack data persistence layer using SQLAlchemy ORM with PostgreSQL, authoring 3 incremental Alembic migrations for schema versioning, and containerized the entire multi-service architecture (agent, server, web) using Docker Compose for consistent local and production deployments.

Nov 2025

Skills

- **Languages:** Python, Java, JavaScript, TypeScript, MySQL, MongoDB, PostgreSQL, Redis, SQLite, C
- **Frameworks:** React, Node.js, Spring Boot, FastAPI, Flask, REST APIs, Microservices, SOA, Tailwind CSS, Next.js
- **Cloud & DevOps:** AWS (Lambda, EC2, S3), Docker, CI/CD, Jenkins, GitHub Actions, Kubernetes
- **Others:** Git, Agile/Scrum, JIRA, Selenium, Pytest, TDD, OAuth 2.0, JWT, Unit/Integration testing, Responsive Web Design, Data Structure